Practical Exercises 1: CopelliaSim

Objective:

In this practical exercise you should learn and apply the basic fundamentals in robotics. For this reason a simulation software is used: CoppeliaSim from Coppelia Robotics. There is a free **educational** version available.

Preparation:

Download the current **educational** version of the software CoppeliaSim from the homepage of Coppelia Robotics GmbH, http://www.coppeliarobotics.com/

Task 1:

Start the program CoppeliaSim and load different so called “scenes” and start the simulation.

A simulation in CoppeliaSim can be started, paused and stopped with

[Menu bar --> Simulation --> Start/Pause/Stop simulation] or through the related toolbar buttons:



[Simulation start/pause/stop toolbar buttons]

Play around with the system, take a look into the CoppeliaSim user manual, which can be accessed by

[Menu bar --> Help --> Help Topics]

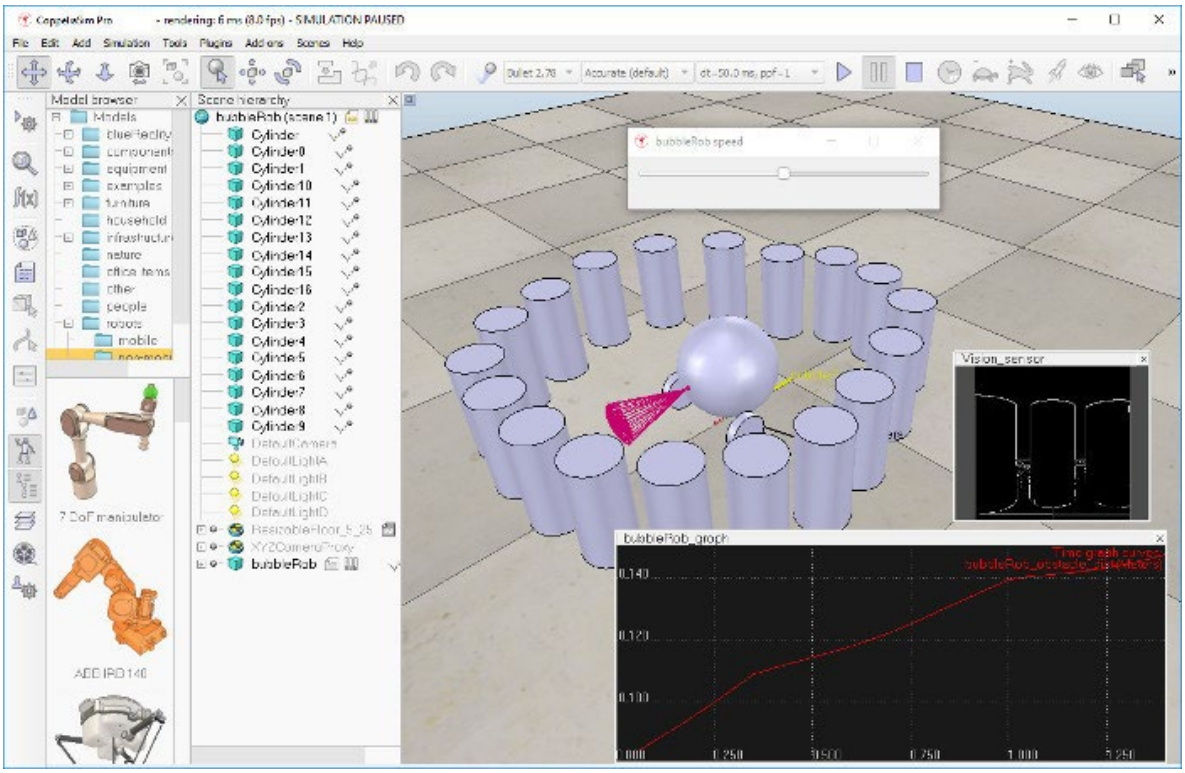
Task 2:

Implement a scene in CoppeliaSim, which is explained in the following BubbleRob tutorial. The scene consists of several obstacles and a two-wheeled robot with a vision and a proximity sensor.

Try to understand the script code at the end, which is written in Lua programming language.
Do several adjustments in the scene by trying things out (changing colors, behaviours, ..).

Building BubbleRob

This tutorial will try to introduce quite many CoppeliaSim functionalities while designing the simple mobile robot *BubbleRob*. The final CoppeliaSim scene file related to this tutorial is located in *scenes/tutorials/BubbleRob*. Following figure illustrates the simulation scene that you will design:



First of all, freshly start CoppeliaSim. The simulator displays a default scene.

You can find the tutorial at [Menu bar --> Help --> User manual]. A web page opens, so you need internet access. Under the link Tutorials you find (among others) the BubbleRob tutorial.

Please note, that this tutorial has been issued with an older version of CoppeliaSim, so you have to expect explanations, which does not fit to the current UI of CoppeliaSim. In that case consult the forum (see just opened web page).

During the this course you will program in python and will connect to CoppeliaSim via an API. This API is described also on this page under the link Writing Code -> CoppeliaSim API framework.

This API is also used in task 3 of this practical exercise.

Task 3 – Remote API

In the next part we will learn how to control the BubbleRob in Coppeliasim with an external Python script over the **ZeroMQ** remote API.

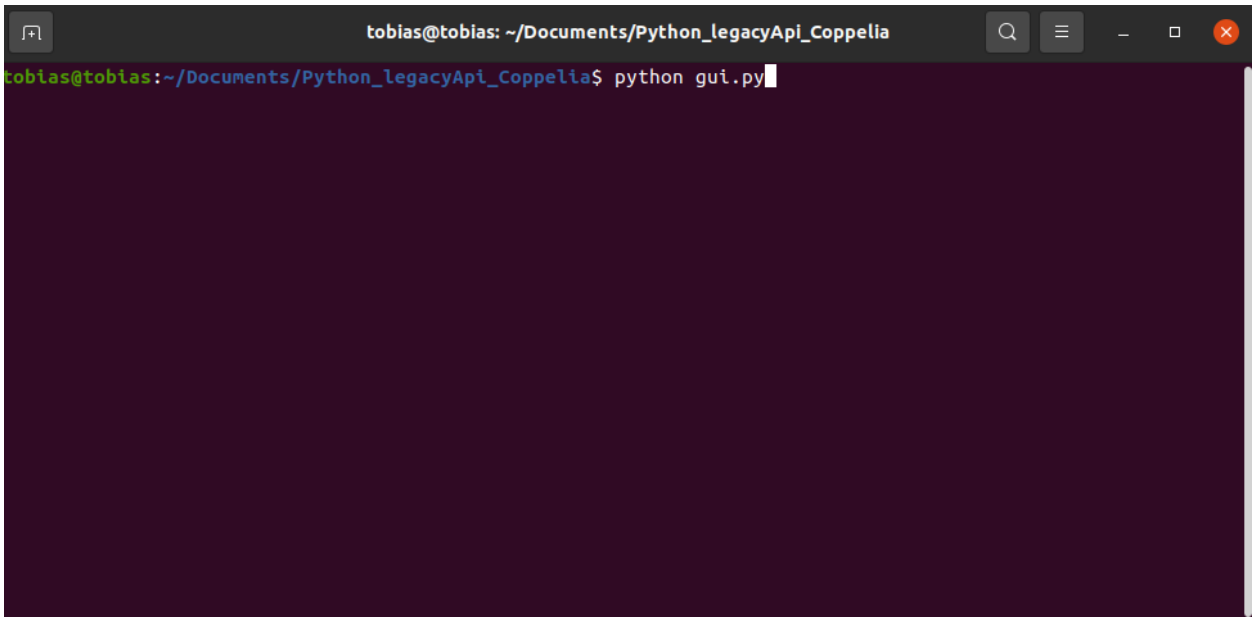
First we save our current status from the tutorial under „bubbleRob.ttt“ [Menu bar --> File --> save scene]. Then we create a copy of that file for our next steps under the name „bubbleRob_GUI.ttt“ [Menu bar --> File --> save scene as --> CoppeliaSim scene…]. Now delete every cylinder this scene so BubbleRob has enough space to drive arround. Next open the **child script** of the BubbleRobs body and delete the code.

Now our scene is prepared for our external Script. For our GUI we need to install the packet **PySide6** with the python packet manager and the packages **ZeroMQ** and **CBOR** for the ZeroMQ API of CoppeliaSim. In addition copy the directory *..\CoppeliaRobotics\CoppeliaSimEdu\programming\zmqRemoteApi\clients\python\zmqRemoteApi\src\coppeliasim_zmqremoteapi_client* into your current working directory.

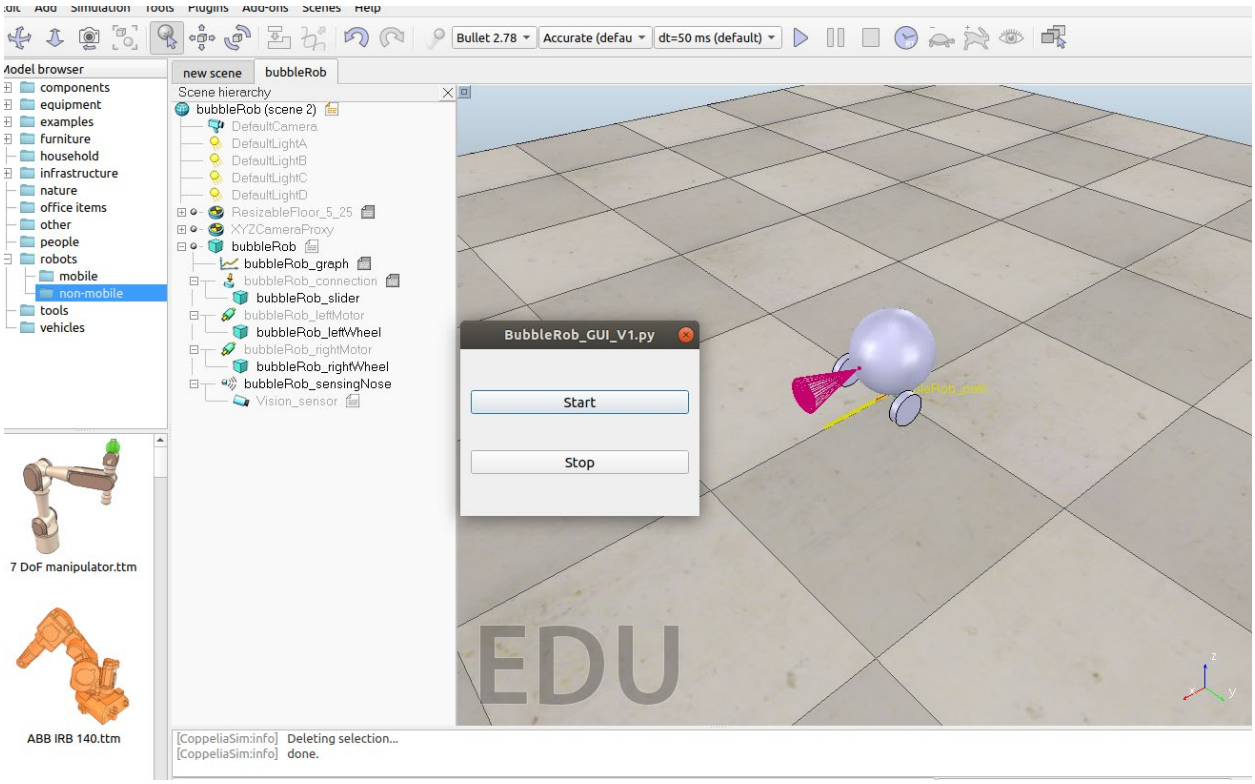
Now open a python code IDE (for example PyCharm) with the provided file gui_1.py.

Open the CoppeliaSim scence bubbleRob_GUI.ttt (if it is still not open), but do not start the simulation.

Run the python script gui_1.py.



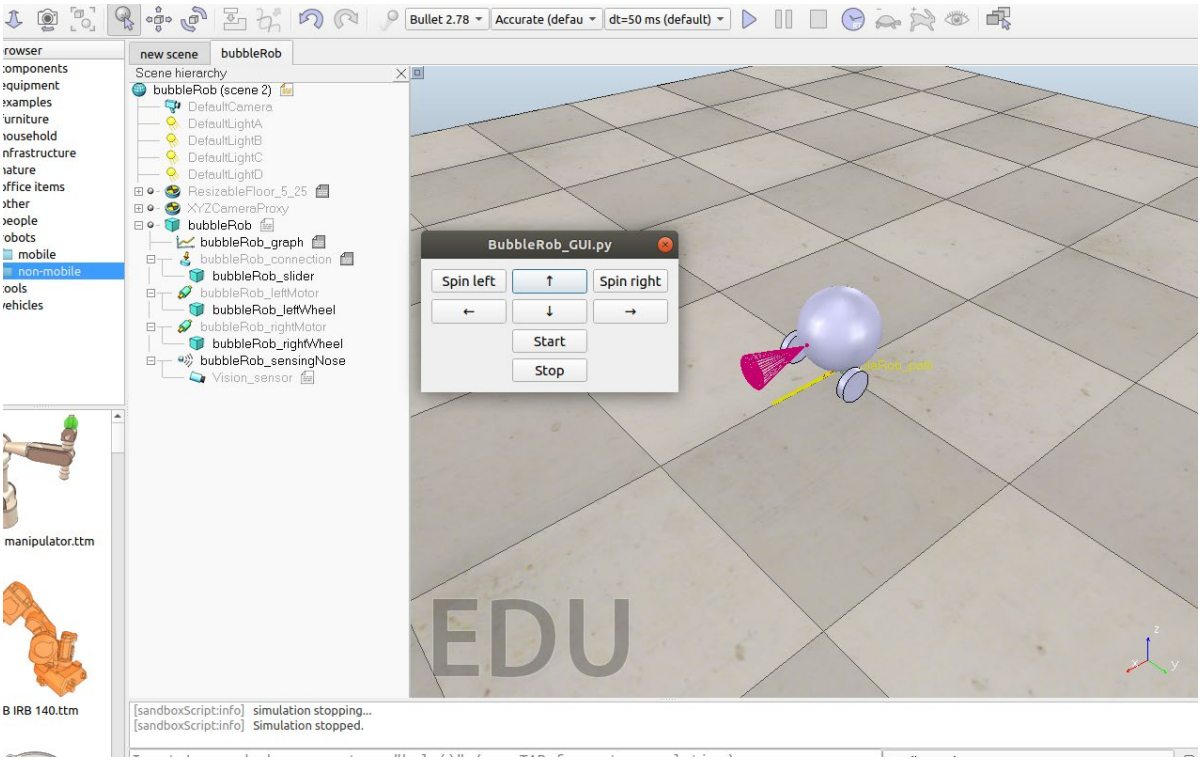
Now you should see a similar GUI:



If you click the start button the simulation should start and run. The BubbleRob motors are at opposite velocities, so it spins around. By clicking the Stop button the BubbleRob stops, since the velocities are set to 0.

Now we add some more functionality by adding a few buttons (see file gui_2.py):

If you run the script you should now have this GUI:



Press the start button, then you can change the direction of the BubbleRob with the arrow buttons. If you press stop the BubbleRob stops. If you exit the GUI the simulation stops and the BubbleRob returns to its original position.

Report:

It is not necessary to write a report.